

Gas Field Policy 27: ISO 13631 (E), August 1, 2002, Comments / Exceptions

<u>Paragraph</u>	<u>Comments</u>
3.7	Siemens Energy defines MAWP as the maximum continuous gauge pressure for which the manufacturer nameplates it's compressors (or relevant part thereof) when handling the specified fluid at the specified temperature. It is also the maximum relief valve setting.
3.10	Siemens Energy defines RDP (rated discharge pressure) as being MAWP/1.10 or MAWP - 25 PSIG (1.8 kg/cm ²) whichever is greater. I.E., a cylinder would have an RDP of 1000 PSIG (70.3 kg/cm ²) and a MAWP of 1100 PSIG (77.3 kg/cm ²) - or a cylinder with a MAWP of 100 PSIG (7.0 kg/cm ²) would have an RDP of 75 PSIG (5.3 kg/cm ²). The MAWP is the controlling rating.
3.11	Siemens Energy Sizing performance program calculates "Rated Power" by assuming some general pressure drops for each of these factors. Each Reseller designs their own pulsation suppression devices, orifices, process piping, intercoolers and separators. Siemens Energy will not be responsible for errors granted via outside suppliers equipment.
3.13	Capacity can be expressed with no negative tolerance if so specified.
3.15	Volume flow referred to 0.1013 Mpa (1.013 barA) and a temperature of 0° C (32° F) if so specified.
6.3	Discharge temperatures can be limited to 135° C (275° F) if so specified. Siemens Energy sizing program flags T (adiabatic) @300° F (149° C) and T (estimated actual) at the cylinder design limit of 350° F (177° C). Both temperatures are shown on standard output form.
6.4.1.2	Siemens Energy calculates gas rod load at the compressor flange at 0 and 180 degrees. Gas rod load can be calculated at relief valve set pressure, if specified.
6.5.1.1	Siemens Energy compressor cylinder "maximum allowable working pressure" (MAWP) exceeds the "rated discharge pressure" (RDP) by 10% or as a minimum 25 PSI (1.8 kg/cm ²).
6.5.2.1	Siemens Energy recommends outboard cylinder supports on all cylinders, 15" (381.0 mm) diameter or larger and all cylinders with HOVVCP.
6.5.2.3.3	Siemens limitation is -51° C (-60° F) before special stud material is required. For temperatures of -20° C (4° F) but warmer than -51° C (-60° F), Siemens Energy can supply special material at an extra price. Note that stainless steel metallic valve plates are required on PF valves for temperatures colder than -29° C (-20° F).
6.5.3.1	Siemens Energy 26.0" (660.4 mm), 26.5" (673.1 mm) and 28.0" (711.2 mm) HOS/HOSS cylinders have a machined rectangular studded flange. Forged steel billet cylinders to 4400 PSI (309.4 kg/cm ²) MAWP typically use "ring joint" flanges. Forged steel billet cylinders from 4400 psi (309.4 kg/cm ²) to 6600 PSI (464.0 kg/cm ²) MAWP use "lens" style connections.
6.6.2.1	Siemens Energy Model PF valves are not 100% polarized. Should a discharge valve inadvertently be installed in a suction valve hole; the valve will not completely seat so no damage can occur.
6.7.1	Siemens Energy standard does not include provision to positively lock piston rods into crossheads.

- 6.7.3. Siemens Energy standard complies with the requirement that combo piston rings or wear bands do not over-travel the valve ports or counter-bores by over 1/2 the axial width of the combo piston ring or wear band.
- 6.7.4. Siemens Energy Model MOS/HOS/HOSS/BOS compressors comply. Siemens Energy Model MOS compressors are not hardened to meet the 50Rc hardness requirement in the packing travel area. If specified, this can be provided at an extra price.
- 6.9.3 Siemens Energy distance piece come standard with a 3/4" (19.1 mm) NPT vent connection. If so specified, a larger connection can be furnished at an extra price.
- 6.10.2 Siemens Energy has a few MOS, HOS and HOSS applications where the packing cases are rated slightly less than the MAWP of the cylinder, however field experience has proven these to be reliable.
- 6.11.2 Siemens Energy standard oil level gauge is a "Bulls Eye" type with a single line only. The Instruction Book defines how to determine minimum and maximum levels with this gauge.
- 6.11.3.1 All Siemens Energy current production line compressor models comply.
- 6.11.3.2 Siemens Energy recommends the use of an electric motor driven pre-lube pump on all electric motor driven compressors. These are normally supplied by the Reseller, but Siemens Energy can supply at an extra price if specified.
- 6.11.6 Siemens Energy supplies a machined connection point for a crankcase oil-level control device. Oil level control device and oil storage tank are normally supplied by the Reseller but can be supplied by Siemens Energy at an extra price if specified.
- 6.11.7 Siemens Energy's long time supplier of lubricator heaters exceeds the 2 W/cm² maximum heat flux specified.
- 6.12.1 Siemens Energy standard is to supply 300 Series stainless steel tubing and fittings for all cylinder lubrication tubing lines.
- 6.12.2 Siemens Energy's standard cylinder lubrication system uses oil supplied from the crankcase to the lubricator pumps and then to the distribution block system. This system can be modified to one supplied from a separate oil tank if specified.
- 6.13.1.3 Siemens Energy has had many years of satisfactory experience with both cast iron and nodular iron cylinders that exceed these ratings. (For example, 10.5" (266.7 mm) CI cylinders to 1500 PSIG (105.5 kg/cm²), 6.0 (152.4 mm) - 7.0" (177.8 mm) NI cylinders to 2750 PSIG (193.3 kg/cm²), 8.5 (215.9 mm) -9.0 (228.6 mm) - 9.5" (241.3 mm) cylinders to 1925 PSIG (135.3 kg/cm²), etc.)
- 6.13.3.3 When used, drilled plugs for plug repair are subject to a liquid dye penetrant examination and tested via hydrotest after repair is complete.
- 7.4.1 Siemens Energy Sizing Performance Program uses a "warning flag" for **discharge** volumetric efficiencies lower than 10% rather than a **suction** volumetric efficiency limit.
- 7.4.2 Siemens Energy does not vent the outboard side of hand operated variable volume clearance pockets as standard. This can be done at an extra price if specified.

- 9.3.3 Siemens Energy offers the Model HOSS compressor using a 2.875" (73.0 mm) piston rod standard design packing cases. Water cooled packing cases can be supplied for high pressure applications or for special gases when specified.
- 17.3.4.1 Siemens Energy can perform a pressure test with helium (in addition to the hydrotest) for gases with a MW of 12 or less or with 0.001 mole fraction of H₂S when specified at an extra price.
- 20.2 Hydrogen Sulfide: Siemens Energy has years of satisfactory experience supplying compressors for H₂S service. General Guidelines are included in GFP18 Sour Gas Service which is included in the Application and Data Book or is available on request. Standard nodular iron cylinder material is ASTM A536. Special materials are available upon request.
- 20.3 Carbon Dioxide: Siemens Energy has years of satisfactory experience supplying compressor for CO₂ service. General Guidelines are included in GFP16 Carbon Dioxide Service which is included in the Application and Data Book or is available upon request.
- 21.8 Siemens Energy's standard primer is Sherwin Williams Red Oxide "Kem-Flash" 500 primer. Top coats supplied by the Reseller need to be compatible with this primer.